Day: 07

* Week 1 Review

Q1) What are the 7 OSI layers and what does each one do?

→ • Layer 1 → Physical Layer : Manages the physics of the network like physical devices.

• Layer 2 → Data Link Layer : Mac Addressin, frame delivery on same network.

• Layer 3 → Network Layer: Logical addressing and routing (IP addressing).

• Layer 4 → Transport Layer: End-to-End delivery, segmentation (TCP, UDP).

• Layer 5 → Session Layer : Mangge sessions (Create, delete)

• Layer 6 → Presentation Layer: Data formatting (SSL / TLS).

• Layer 7 → Application Layer : What users see (HTTP, HTTPS, DNS).

Q2) What happens during a TCP 3-way handshake, packet by packet? Write the flags and sequence numbers.

→ • Step 1: SYN (Client → Server)
- Flags : SYN = 1
- Client ISN (Initial Sequence Number): eg: 100

- Meaning: "I want to connect, starting at seq 100".

- Step 2: SYN - ACK (Server → Client)

- Flags: SYN = 1, ACK = 1
- Server ISN: eg: Seq = 300, ACK = 101 (Client seq + 1)
- Meaning: "Acknowledged, I'll start at Seq 300".

- Step 3: ACK (Client → Server)

- Flags: ACK = 1
- Seq = 101, ACK = 301 (Server's seq + 1)
- Meaning: "Got it. Connection Established"

- After this data transfer begins.

Q3) What is a DNS A record and what does it return?

→ • A DNS A record maps a domain name → IPv4 address.

• Returns: A 32-bit IPv4 address (eg: google.com → 142.250.195.46)

• It's the most fundamental DNS record type used for hostname resolution.

• Eg: nslookup example.com → returns 93.184.216.34

Q4> What is the difference between HTTP and HTTPS in Wireshark? What can you see in each?

→

Feature	HTTP	HTTPS
Port	80	443
Wireshark Filter	http	tls or ssl
Payload visible	Yes fully	No - encrypted
What you can see	Full request/response, headers, cookies, POST body URLs.	IP addresses, ~~ports~~, TLS handshake, certificate info, SNI
Protocol Shown	HTTP	TLSv1.2/TLSv1.3

Q5> What is SNI?

→ SNI is a TLS extension sent in the Client Hello packet that tells the server which hostname the client is trying to reach — before any encryption begins